Advanced Calculus I Non-hand-in Assignment 2

This homework comes from textbook and will **NOT** be collected (這份作業完成後不需要繳交、評分。). But for practice,or quizzes or exams, please make sure that you have idea to solve the following questions by yourself. (這些題目是很重要的練習，小考、期中、期末考會有類似的題目。練習這些題目的過程中所逐漸形成我們對於所學到的知識的"看法"，會在考試中很有幫助。所以，請一定要了解如何解題與解題背後的想法是什麼。)

Please prove the following propositions are true. (請證明下列命題爲眞。)

Section 1.3.
1.11 Example (on page 30)
1.13 Remark (on page 31)
1.18 Theorem (on page 33)

第 2 週｜自主練習・課本題目補充

William R. Wade | An Introduction to Analysis | Fourth Edition, Global Edition

依教師 Non-hand-in Assignment 指定順序整理；以下為題目／命題敘述，不含解答。

Example 1.11

Wade p.30

1.11 ***EXAMPLE.***

If $E = [0, 1]$, prove that $\sup E = 1$.

Remark 1.13

Wade p.31

1.13 Remark. *If a set has a supremum, then it has only one supremum.*

Theorem 1.18

Wade p.33

1.18 Theorem. [DENSITY OF RATIONALS].
If $a, b \in \mathbf{R}$ satisfy $a < b$, then there is a $q \in \mathbf{Q}$ such that $a < q < b$.